

Changes in signalling pathways regulating protein synthesis in human muscle in the recovery period after endurance exercise

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Abstract

Aim: Exercise induced alterations in the rate of muscle protein synthesis may be related to activity changes in signalling pathways involved in protein synthesis. The aim of the present study was to investigate whether such changes in enzyme phosphorylation occur after endurance exercise.

Methods: Six male subjects performed ergometer cycling exercise for 1 h at 75% of the maximal oxygen uptake. Muscle biopsy samples from the vastus lateralis were taken before, immediately after, 30 min, 1 h, 2 h and 3 h after exercise for the determination of protein kinase B (PKB/Akt), mammalian target of rapamycin (mTOR), glycogen synthase 3 kinase (GSK-3), p70S6 kinase (p70^{S6k}) and eukaryotic elongation factor 2 (eEF2) phosphorylation.

Results: The phosphorylation of Akt was unchanged directly after exercise, but two- to fourfold increased 1 and 2 h after the exercise, whereas GSK-3 α and β phosphorylation were two- to fourfold elevated throughout most of the 3-h recovery period. Phosphorylation of mTOR was elevated threefold directly after, 30 min and 2 h after exercise and eEF2 phosphorylation was decreased by 35–75% from 30 min to 3 h-recovery. Exercise led to a five- to eightfold increase in Ser⁴²⁴/Thr⁴²¹ phosphorylation of p70^{S6k} up to 30 min after exercise, but no change in Thr³⁸⁹ phosphorylation.

Conclusions: The marked decrease in eEF2 phosphorylation suggests an activation of translation elongation and possibly protein synthesis in the recovery period after sustained endurance exercise. The lack of p70^{S6k} activation suggests that translation initiation is activated via alternative pathways, possibly via the activation of eukaryotic initiating factor 2B.

Keywords GSK-3, mTOR, p70S6 kinase, translation initiation.

Exercise training leads to adaptations in the skeletal muscle that depends on the type of work the muscle performs. Resistance exercise will increase the muscle strength and mass, whereas endurance exercise involves increased vascularization, mitochondrial density and oxidative capacity, leading to an increase in fat oxidation and endurance (Holloszy & Booth 1976, Henriksen 1977, Saltin & Gollnick 1983). Short *et al.* (2004)

showed that muscle protein synthesis at rest was increased by 22% after a 16-week endurance exercise training programme and that this effect was not age or gender-specific. Picosky *et al.* (2006) reported an increase in muscle protein fractional synthetic rate and net protein balance after 4 weeks of aerobic training. However, only a few studies have examined the acute effects of a single bout of endurance exercise on muscle

protein synthesis in human subjects. Using the stable isotope technique, protein synthesis was reported to be unaltered in the skeletal muscle of young healthy subjects during 4 h of treadmill walking (Carraro *et al.* 1990), but significantly increased in the recovery period following 45 min or 4 h of exercise (Carraro *et al.* 1990, Sheffield-Moore *et al.* 2004). At a higher intensity, 1 h of one-legged kicking exercise at 67% of the maximal work rate, the fractional rate of protein synthesis was elevated from 6 to 48 h after exercise, with a peak at 24 h (Miller *et al.* 2005). In contrast, Tipton *et al.* (1996) found no increase in protein synthesis in female swimmers after high-intensity swimming.

The type of exercise and the intensity and duration of the exercise are probably critical factors in the regulation of protein synthesis. The results provide some evidence of a temporary increase in protein synthesis in the recovery phase after a single bout of sustained endurance exercise and increased protein synthesis capacity after endurance exercise training. The molecular mechanisms behind this adaptation are not known, but is likely to be mediated through phosphorylation and activation of enzymes that regulate translation initiation. Many studies have linked the phosphorylation of protein kinase B (PKB or Akt) and the mammalian target of rapamycin (mTOR) to increased protein synthesis in myotubes subjected to mitogenic stimulation by insulin (Kimball *et al.* 2002) and IGF-1 (Rommel *et al.* 2001) as well as compensatory hypertrophy in rat muscle (Bodine *et al.* 2001). Furthermore, resistance exercise in rats, as well as high-frequency stimulation of skeletal muscle *in vitro*, has also been reported to increase Akt and mTOR phosphorylation (Bolster *et al.* 2003, Atherton *et al.* 2005). Activation of mTOR leads to phosphorylation and activation of p70S6 kinase (p70^{S6k}), thereby increasing the translation of mRNA that encodes for ribosomal proteins and elongation factors (Jefferies *et al.* 1997, Kimball & Jefferson 2006). In line with this, Dreyer *et al.* (2006) recently reported that heavy resistance exercise led to increases in Akt, mTOR and p70^{S6k} phosphorylation and a decrease in phosphorylation of eukaryotic elongation factor 2 (eEF2) in the 2-h recovery after resistance exercise in human subjects. These changes occurred in parallel with an increase in the rate of protein synthesis. In contrast, low-frequency stimulation of the skeletal muscle *in vitro* or treadmill running did not influence Akt, mTOR or p70^{S6k} phosphorylation in rat muscle (Nader & Esser 2001, Atherton *et al.* 2005). Such measurements have not yet been performed in human muscle after endurance exercise.

In the present study, the effect of sustained endurance exercise on the phosphorylation of Akt, GSK-3, mTOR, p70^{S6k} and eEF2 in the recovery period after

exercise was investigated. Skeletal muscle biopsy samples were taken before and repeatedly during a 3-h recovery period, and the kinase phosphorylation was determined by immunoblot analyses using phosphospecific antibodies. In addition, muscle glycogen was determined and blood samples were taken for the measurements of insulin, glucose and lactate concentrations. The results suggest a temporary increase in the activation of proteins that may stimulate protein synthesis in the recovery phase after endurance exercise.

Materials and methods

Subjects

Six healthy men, mean $\pm$ SE, age 23 ± 1 years, height 182 ± 3 cm, weight 72 ± 2 kg and maximal oxygen uptake ($\text{VO}_{2\text{max}}$) 3.86 ± 0.16 L min^{-1} participated in the study. The subjects reported performing endurance exercise training less than twice weekly. They were fully informed of the nature and the possible risks associated with the study before they volunteered to participate. The study was approved by the Ethics Committee at the Karolinska Institute.

Preliminary tests

All exercise tests were performed on a computerized cycle ergometer (Monark 839E; Monark, Varberg, Sweden). Prior to the experiment, the subjects participated in two preparatory tests. The first one was designed to determine the oxygen uptake of the subjects at three submaximal work rates along with their $\text{VO}_{2\text{max}}$ by the use of an on-line system (Amis 2001 Automated Metabolic Cart; Innovision A/S, Odense, Denmark). The second preparatory test was conducted to familiarize the subjects with the experimental procedure; the subjects exercised for 45 min at 75% of $\text{VO}_{2\text{max}}$, the same work rate as in the following test exercise. The preliminary tests were carried out at least 1 week prior to the day of the experiment.

Experimental protocol

The subjects were asked to refrain from physical activities and alcohol consumption for 2 days prior to the day of the experiment. They reported to the laboratory in the morning after fasting overnight (since 8 pm the night before). A catheter was inserted into the antecubital vein for blood sampling, and, thereafter, the subjects rested in a supine position for 30 min in order to standardize the testing procedure. A blood sample was taken after 30 min of rest and a biopsy specimen was obtained from the vastus lateralis muscle approxi-

mately 13 cm from the mid patella using a Weil-Blakesley conchotome (AB Wisex, Mölndal, Sweden), as described by Henriksson (1979). The subjects performed a warm-up exercise for 5 min at 100 W on the cycle ergometer, followed by exercise at a work rate demanding 75% of $\dot{V}O_{2\max}$ for 1 h. The oxygen consumption was measured at 15 and 45 min of exercise. After the measurement at 15 min of exercise, the work rate was adjusted, if necessary, to correspond to 75% of $\dot{V}O_{2\max}$. The heart rate was continuously monitored throughout the exercise period, using a Polar S610 heart rate monitor (Polar Electro Oy, Kempele, Finland). The subjects were allowed to drink water *ad libitum* during the exercise. Blood samples were collected after 20, 40 and 60 min of exercise (i.e. during the last min of exercise), and in the recovery period at 15, 30, 60, 90 and 120 min after exercise. No blood samples were taken during the third hour of recovery. Muscle biopsy specimens were taken immediately after the exercise and during the recovery at 30, 60, 120 and 180 min after the exercise. All tissue samples were obtained while the subjects rested in a supine position. The first, third and fifth biopsy specimens were taken from the right vastus lateralis muscle and the second, fourth and sixth biopsy specimens were taken from the left vastus lateralis muscle. Each biopsy sample was taken from a new incision; the second and third biopsy specimen in each leg was taken approximately 3–4 cm proximal to the site of the previous specimen. The specimens were frozen in liquid nitrogen within 10 s of the withdrawal.

Muscle analysis

Muscle biopsy samples were freeze-dried and dissected free from blood, connective tissue and adipose tissue. The glycogen content was determined in 1–2 mg of the dissected muscle according to the method described by Leighton *et al.* (1989). For immunoblot analyses, approximately 2 mg of the samples were homogenized in ice-cold buffer (20 mM Hepes, pH 7.4, 1 mM EDTA, 5 mM EGTA, 10 mM $MgCl_2$, 50 mM β -glycerophosphate, 1 mM Na_3VO_4 , 2 mM DTT, 1% Triton X-100, 20 $\mu g\ mL^{-1}$ leupeptin, 50 $\mu g\ mL^{-1}$ aprotinin, 40 $\mu g\ mL^{-1}$ PMSF) at a dilution of 160 $\mu L\ mg^{-1}$ of dry weight muscle. Homogenates were rotated for 60 min at 4 °C and centrifuged at 10 000 g for 10 min at 4 °C. The protein concentration was determined in aliquots of the supernatant diluted 1 : 10 in ultrapure water (Millipore, Billerica, MA, USA) using a BCA protein assay (Pierce Biotechnology, Rockford, IL, USA).

The aliquots of the muscle lysate were mixed with the Laemmli sample buffer and heated in order to denature proteins. Samples containing 40 μg total protein were

separated by SDS-PAGE using 4–20% gels on a Mini Protean 3 or Criterion electrophoresis cell (Bio-Rad Laboratories, Richmond, CA, USA). Following electrophoresis, gels were incubated in transfer buffer (25 mM Tris-HCl, 192 mM glycine and 20% methanol) for 25 min in order to equilibrate gels for optimal transfer. Proteins were then transferred to polyvinylidene fluoride (Akt, eEF2, p70^{S6k} and mTOR at Ser²⁴⁸¹) or nitrocellulose (GSK-3 and mTOR at Ser²⁴⁴⁸) membranes (Bio-Rad Laboratories) at 100 mA constant current for 2 h (Akt and GSK-3), 3 h (p70^{S6k} and mTOR) or 14 h (eEF2) on ice in a cold room at 4 °C. Membranes were blocked in TBS (10 mM Tris pH 7.6, 100 mM NaCl) containing 5% non-fat dry milk (eEF2, GSK-3 at Ser^{21/9}, p70^{S6k} at Ser⁴²⁴/Thr⁴²¹ and mTOR) or in StartingBlock blocking buffer (Akt and p70^{S6k} at Thr³⁸⁹) (Pierce Biotechnology) for 1 h and then incubated for 2 h at room temperature (eEF2 and p70^{S6k} at Ser⁴²⁴/Thr⁴²¹) or overnight at 4 °C (Akt, p70^{S6k} at Thr³⁸⁹ and mTOR) with primary phosphospecific polyclonal antibodies; Akt at Ser⁴⁷³, eEF2 at Thr⁵⁶, GSK-3 at Ser^{21/9}, p70^{S6k} at Ser⁴²⁴/Thr⁴²¹, mTOR at Ser²⁴⁴⁸ and Ser²⁴⁸¹ (Cell Signaling Technology Inc., Beverly, MA, USA; lot nos 9271L, 2331S, 9331S, 9204S, 2971S and 2974L, respectively) and p70^{S6k} at Thr³⁸⁹ (Santa Cruz Biotechnology, Santa Cruz, CA, USA; lot no K239). Antibodies were diluted 1 : 1000 (Akt, GSK-3, p70^{S6k} at Ser⁴²⁴/Thr⁴²¹ and mTOR) or 1 : 2000 (eEF2 and p70^{S6k} at Thr³⁸⁹) either in TBS with 0.1% Tween 20 (TBS-T) containing 2.5% non-fat dry milk (eEF2, GSK-3, p70^{S6k} at Ser⁴²⁴/Thr⁴²¹ and mTOR) or in StartingBlock blocking buffer containing 0.1% Tween 20 (Akt and p70^{S6k} at Thr³⁸⁹). Membranes were then washed in TBS-T (Akt and p70^{S6k} at Thr³⁸⁹) or TBS-T containing 2.5% non-fat dry milk (eEF2, p70^{S6k} at Ser⁴²⁴/Thr⁴²¹ and mTOR), incubated for 1 h with secondary antibody (horseradish peroxidase-linked anti-rabbit IgG; Cell Signaling Technology Inc.) diluted 1 : 10 000 followed by washing in TBS-T or TBS-T containing 2.5% non-fat dry milk. Phosphorylated proteins were visualized by enhanced chemiluminescence reagents according to the manufacturer's protocol (SuperSignal west femto maximum sensitivity substrate; Pierce Biotechnology) and quantified by densitometric scanning using a Gel Doc 2000, in combination with Molecular Analyst software version 2.1 (Bio-Rad Laboratories). The methods are modified from Markuns *et al.* (1999) and Karlsson *et al.* (2004).

The membranes described above were incubated in RestoreTM Western blot stripping buffer (Pierce Biotechnology) for 30 min at 37 °C and re-probed with appropriate polyclonal antibodies for the detection of total expression levels of each protein, Akt, eEF2 and GSK-3 (Cell Signaling Technology Inc.) and mTOR and p70^{S6k} (Santa Cruz Biotechnology).

Blood analysis

Blood samples were taken from the venous catheter and separated in two glass tubes: one heparinized for glucose and lactate measurements and one containing EDTA for measurements of insulin. The tubes were centrifuged (9000 g for 2 min) and plasma was stored at -80°C . Plasma glucose and lactate concentrations were analysed according to Bergmeyer (1974). The plasma insulin concentration was measured using a radioimmunoassay kit according to the manufacturer's protocol (Linco Research, Inc., St Charles, MO, USA).

Statistics

All data are presented as mean $\pm$ SE of the means. A one-factorial (time) repeated measures ANOVA was employed to analyse changes in glycogen and in the phosphorylation state of the different kinases in muscle samples during and after exercise. When a significant overall effect was indicated, a Fisher LSD *post hoc* test was performed. Significance was accepted at $P < 0.05$.

Results

All the subjects completed the 1-h cycling exercise. The average work rate was 201 ± 15 W, demanding an oxygen uptake of 2.88 ± 0.15 L min^{-1} corresponding to $74 \pm 1.0\%$ of the subject's $\text{VO}_{2\text{max}}$. The average heart rate during the exercise for the six subjects was 171 ± 3 beats min^{-1} , corresponding to $91 \pm 1.4\%$ of their maximal heart rates.

The muscle glycogen content decreased from 381 ± 28 to 84 ± 14 mmol kg^{-1} dw ($P < 0.05$) during exercise and increased significantly during recovery to 134 ± 23 mmol kg^{-1} dw at 3 h after exercise (Table 1). Figure 1 illustrates changes in plasma insulin, glucose and lactate during the exercise. Plasma insulin levels decreased from 9.93 ± 1.24 to 4.14 ± 0.35 mU L^{-1} ($P < 0.05$) during the cycling exercise, and increased 15 min after exercise, but decreased again and remained at approximately 30% below the resting value throughout the recovery period. The glucose concentration decreased from 6.07 ± 0.34 to 5.31 ± 0.32 mmol L^{-1}

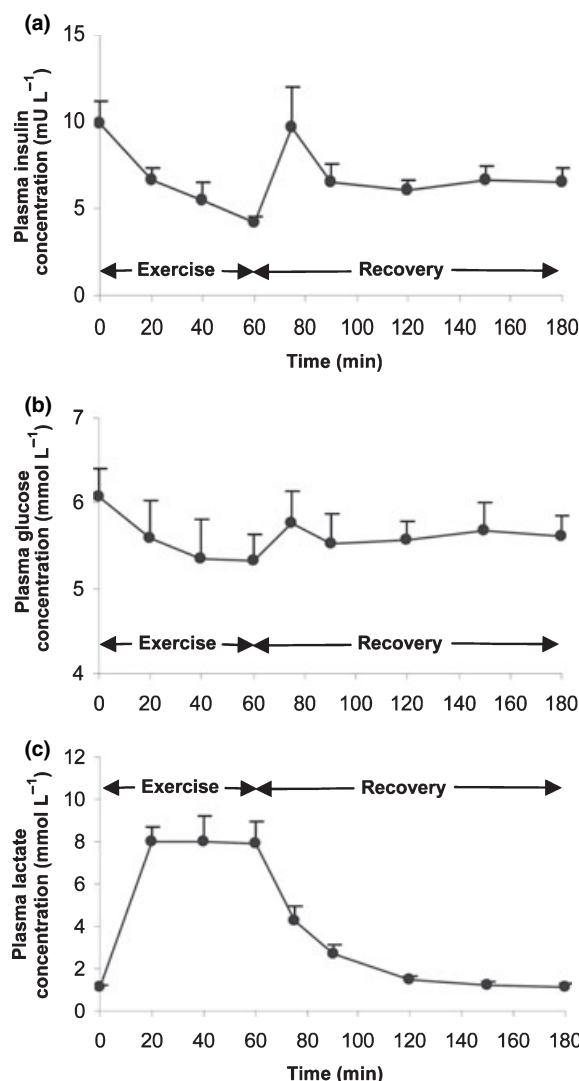


Figure 1 Plasma insulin (a), glucose (b) and (c) lactate concentrations during and after 1 h of cycling exercise. Values are mean $\pm$ SE for six subjects.

($P < 0.05$) during exercise and increased slightly to 5.60 ± 0.25 mmol L^{-1} at 2 h of recovery. The plasma lactate concentration rose from 1.11 ± 0.11 to 8.03 ± 0.68 mmol L^{-1} ($P < 0.05$) after 20 min of exercise and remained at this level throughout the exercise.

Glycogen concentration [mmol (kg dry wt)⁻¹]

Before exercise	After exercise	Post-exercise			
		30 min	1 h	2 h	3 h
381 ± 28	$84 \pm 14^*$	$94 \pm 8^*$	$103 \pm 19^*$	$112 \pm 13^*$	$134 \pm 23^{*\dagger}$

Values are given as mean $\pm$ SE for six subjects.

* $P < 0.05$ vs. before exercise and $^\dagger P < 0.05$ vs. after exercise.

Table 1 Muscle glycogen concentration before, immediately after exercise and repeatedly during the recovery period after 1 h of cycling exercise at 75% $\text{VO}_{2\text{max}}$

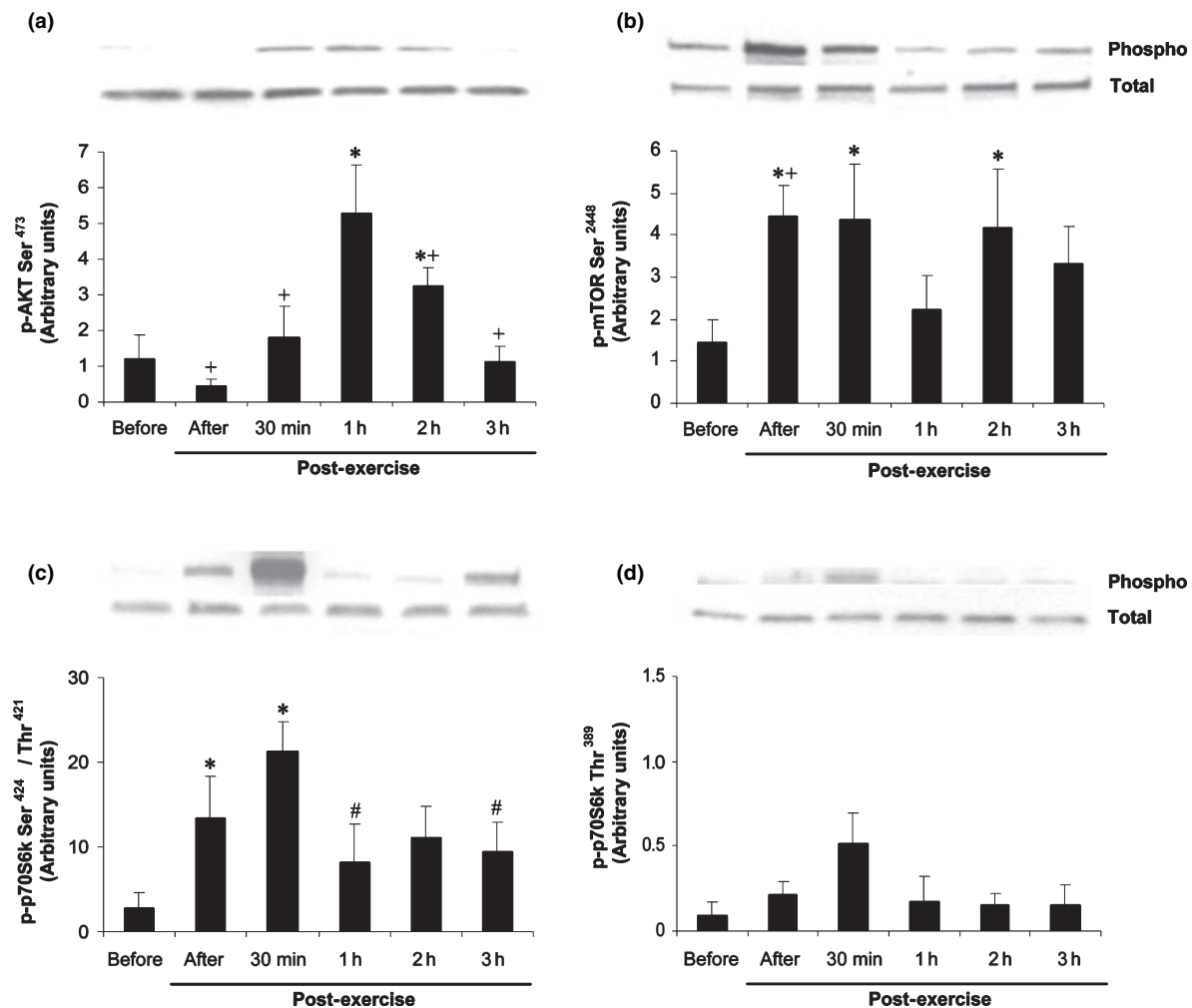


Figure 2 Phosphorylation of Akt on Ser⁴⁷³ (a), mTOR on Ser²⁴⁴⁸ (b) and p70^{S6k} on Ser⁴²⁴/Thr⁴²¹ (c) and Thr³⁸⁹ (d) in skeletal muscle before, immediately after and 30 min, 1 h, 2 h and 3 h after 1 h of cycling exercise. Representative immunoblots are shown at the top of each figure. Values in the graphs are arbitrary units (mean $\pm$ SE for six subjects). * P < 0.05 vs. before exercise, # P < 0.05 vs. 30 min post-exercise and + P < 0.05 vs. 1 h post-exercise.

Akt phosphorylation was not altered immediately after the exercise, but at 1 and 2 h after the exercise a two- to fourfold increase was found, which returned to the initial level 3 h post-exercise (Fig. 2a). The phosphorylation of mTOR at Ser²⁴⁴⁸ was increased threefold directly after exercise and 30 min and 2 h in the recovery period (Fig. 2b), whereas phosphorylation of mTOR at Ser²⁴⁸¹ was unaffected at all time points (data not shown). Directly after the exercise, a five- to eightfold rise in Ser⁴²⁴/Thr⁴²¹ phosphorylation of p70^{S6k} was detected that persisted for 30 min into the recovery period, whereas phosphorylation at position Thr³⁸⁹ did not change significantly after the exercise (Fig. 2c,d). The elongation factor eEF2 decreased its phosphorylation at Thr⁵⁶ by 60% at 30 min after exercise, reached its lowest value 1 h after the exercise and increased again between 1 and 3 h of the recovery, but was still 35% lower at 3 h after the exercise as

compared to the resting value (Fig. 3a). GSK-3 α phosphorylation at Ser²¹ was three- to fourfold elevated from 30 min to 3 h after the exercise and peaked at 1 h (Fig. 3b). The GSK-3 β phosphorylation at Ser⁹ was two- to threefold increased at all time points after exercise (Fig. 3c).

Discussion

The novel findings in this study are that an acute bout of endurance exercise leads to increases in the phosphorylation of mTOR at Ser²⁴⁴⁸ and decreases in the phosphorylation of eEF2 at Thr⁵⁶. The phosphorylation of mTOR was elevated up to 2 h after the exercise, whereas eEF2 was reduced from 30 min to 3 h of recovery. None of these enzymes have been previously investigated in the human muscle in the recovery period after endurance exercise. The phosphorylation of Akt

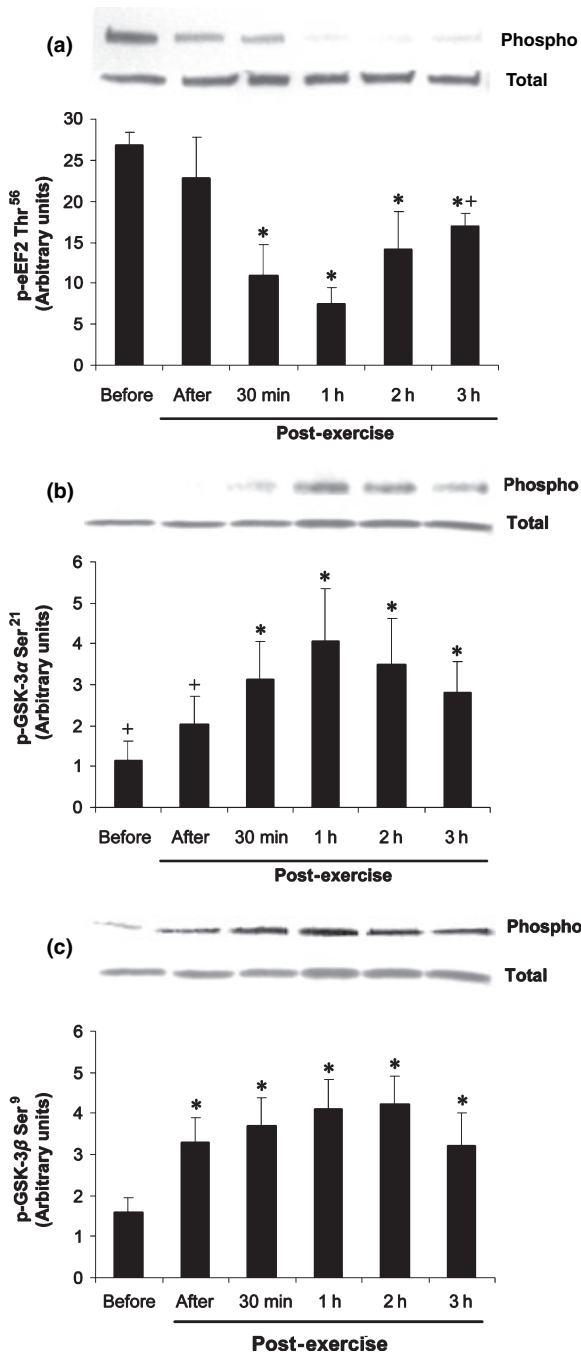


Figure 3 Phosphorylation of eEF2 on Thr⁵⁶ (a), GSK-3α on Ser²¹ (b) and GSK-3β on Ser⁹ (c) in skeletal muscle before, immediately after and 30 min, 1 h, 2 h and 3 h after 1 h of cycling exercise. Representative immunoblots are shown at the top of each figure. Values in the graph are arbitrary units (mean ± SE for six subjects). **P* < 0.05 vs. before exercise and +*P* < 0.05 vs. 1 h post-exercise.

on Ser 473 remained unchanged immediately after exercise, but increased 1 and 2 h after exercise and had returned to the basal level 3 h after exercise. The different time course for the changes in Akt and mTOR

phosphorylation (i.e. the much earlier phosphorylation of mTOR) suggests that mTOR is activated not only by Akt but also via alternate pathways. An Akt independent increase in mTOR phosphorylation was also found in human muscle after the resistance exercise in combination with supply of branched-chain amino acids (Blomstrand *et al.* 2006) and in rat muscle stimulated to contract (Parkington *et al.* 2003).

The effect of exercise on p70^{S6k} was limited to Ser⁴²⁴/Thr⁴²¹, as phosphorylation of Thr³⁸⁹ remained unaltered, indicating that this kinase was not activated (Pearson *et al.* 1995). In agreement with this, Coffey *et al.* (2006) found that an acute bout of aerobic exercise did not activate p70^{S6k} in humans evaluated from the lack of an effect on the ribosomal protein S6. Similar findings were found in studies on the rodent skeletal muscle, where p70^{S6k} activity is increased in the skeletal muscle only in response to either high force isometric or eccentric contractions, but not after contractions produced by low-frequency electrical nerve stimulation or treadmill running (Nader & Esser 2001, Haddad & Adams 2002, Atherton *et al.* 2005). The lack of p70^{S6k} phosphorylation despite an increase in mTOR phosphorylation suggests another role for mTOR following endurance exercise. There is some evidence that mTOR regulates translation elongation via phosphorylation of eukaryotic elongation factor 2 kinase (eEF2k) at several sites and thereby deactivating the kinase (Proud *et al.* 2001, Browne & Proud 2002, 2004). One of these sites is a direct target for p70^{S6k}, whereas the kinases acting at the other sites have not yet been identified (Wang & Proud 2006). Inactivating eEF2k causes a dephosphorylation of eEF2 at Thr⁵⁶ and eEF2 becomes activated (Redpath *et al.* 1993). In opposite, when phosphorylated at this residue, eEF2 becomes inactive and is not able to attach to ribosomes and thereby inhibit the most energy-consuming part of the translation (Browne & Proud 2002). Recently, Rose *et al.* (2005) found that eEF2 phosphorylation increased during cycling exercise suggesting a decrease in elongation during the actual exercise. In the present study, eEF2 was dephosphorylated in the recovery period after exercise, which may indicate that translation elongation is activated and protein synthesis rate increased. In support of this notion is the observation that muscle protein synthesis increased in association with a decrease in the phosphorylation of eEF2 after resistance exercise (Dreyer *et al.* 2006). One possible regulator of the eEF2 activation after exercise may be the deactivation of eEF2k via mTOR. Another possible regulatory mechanism could be via the mitogen-activated protein kinases (MAPK) ERK1/2 and p38, which have been implicated as regulators of the eEF2k (Knebel *et al.* 2001, Wang *et al.* 2001). In the ERK-pathway the effect seems to be mediated through activation of the

downstream target p90 ribosomal S6 kinase (p90rsk). In fact, an activation of the MAPK signalling cascade, including a 3-fold increase in the activity of p90rsk, was found after a marathon race (Yu *et al.* 2001), suggesting a possible role for p90rsk in the regulation of eEF2 kinase after sustained endurance exercise.

The lack of effect on Akt phosphorylation directly after exercise (Fig. 2a) is in agreement with some studies on endurance exercise (Widegren *et al.* 1998, Wojtaszewski *et al.* 2001), but is in contrast to others (Thorell *et al.* 1999, Sakamoto *et al.* 2004, Deshmukh *et al.* 2006, Wilson *et al.* 2006). The reason for the divergent results is not clear; differences in duration and intensity of the exercise are probably important factors as well as the training history of the subjects. Coffey *et al.* (2006) found an increase in Akt phosphorylation after endurance exercise only in endurance-trained subjects; no effect was observed in strength-trained individuals. Furthermore, insulin is a potent activator of Akt and a decrease during the exercise could, therefore, work against a contraction-induced activation as suggested by Sakamoto *et al.* (2004). In the present study, the insulin level was reduced by 60% at the end of the exercise and in fact, a similar reduction was found in Akt phosphorylation, but the decrease did not reach statistical significance. In the early recovery, there was a rapid increase in insulin, but a slower rise in Akt phosphorylation, reaching significance at 1 and 2 h after exercise. After 3-h recovery, Akt phosphorylation was similar to the basal level, which is in agreement with the results by Wojtaszewski *et al.* (2001), however, the transient increase found after 1 h and 2 h recovery (Fig. 2a) has not previously been reported.

An increase in Akt activation may be responsible, at least in part, for inhibiting both the alpha and the beta isoform of GSK-3 when phosphorylated at Ser²¹ and Ser⁹, respectively (Cross *et al.* 1995). However, the data also suggest that other factors are involved in the regulation of GSK-3 activity as a rapid phosphorylation of both isoforms occurred; GSK-3 α was phosphorylated at 30 min after, and GSK-3 β directly after exercise. Akt-independent phosphorylation or deactivation of GSK-3 was previously found in studies on the rodent muscle during treadmill running or nerve stimulation of the muscle (Markuns *et al.* 1999, Sakamoto *et al.* 2002). The reported effects of exercise on GSK-3 phosphorylation or deactivation in the human muscle vary; Sakamoto *et al.* (2004) found a decrease in activity of GSK-3 α , but no change in the beta isoform after endurance exercise. Wojtaszewski *et al.* (2001) reported a slight increase in GSK-3 α activity but no change in GSK-3 β activity following endurance exercise. In the present study, both the alpha and beta isoforms of GSK-3 were phosphoryl-

ated up to 3 h in the recovery period suggesting an inhibition of both isoforms. The reduction in muscle glycogen during the exercise was larger and the level at the end of the exercise considerably lower in the present study than in the previous studies (Wojtaszewski *et al.* 2001, Sakamoto *et al.* 2004). During the 3-h recovery period, the glycogen content increased from 84 to 134 mmol kg⁻¹ dw (Table 1), which could be related to the activation of glycogen synthase, at least in part, through the inhibition of GSK-3 (Cohen *et al.* 1997, Cross *et al.* 1997, Sakamoto *et al.* 2004) as suggested by the sustained increase in GSK-3 phosphorylation in the recovery period (Fig. 3b,c).

Inactivation of GSK-3 could also be related to increased translation initiation and protein synthesis via phosphorylation and activation of the eukaryotic initiating factor 2B (eIF2B) (Welsh *et al.* 1998, Jefferson *et al.* 1999, Wang *et al.* 2001). Rommel *et al.* (2001) suggested that inhibiting GSK-3 might contribute to skeletal muscle hypertrophy, which was later confirmed by the findings that inhibition of GSK-3 β stimulated hypertrophy in myotubes (Vyas *et al.* 2002). Endurance exercise, like in the present study, is not likely to cause any marked hypertrophy, however, it is tempting to suggest that an increased rate of mitochondrial protein synthesis is mediated also via the exercise-induced inhibition of GSK-3 activity promoting translation initiation via activation of eIF2B. The observed decrease in the eEF2 phosphorylation, given that the phosphorylation activates eEF2, further supports that the rate of protein synthesis is elevated in the 3-h recovery period after the exercise.

In the present study, the subjects exercised in a fasted state without nutritional supply in the recovery period. Endurance exercise in a fed state, or exercise in combination with nutritional supply after the exercise, may have a greater impact on protein signalling because of a combined stimulatory effect of the exercise and nutrition. A prolonged effect on Akt and p70^{S6k} phosphorylation has, in fact, been reported after a dynamic shortening or lengthening exercise, when subjects were given carbohydrates and amino acids in the period up to 24 h after exercise (Cuthbertson *et al.* 2006). However, similar studies in relation to the endurance exercise have not yet been performed.

In conclusion, the present data show that a bout of moderate to high-intensity endurance exercise led to increases in mTOR, Akt and GSK-3 phosphorylation and a decrease in eEF2 phosphorylation in the 3-h recovery period after the exercise. Even though further studies are needed to clarify the regulation of these pathways in response to a sustained endurance exercise, the observed changes in molecular signalling suggest an increase in elongation and possibly protein synthesis in the early recovery phase.

Conflict of interest

There are no conflicts of interest.

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